

Spatio-Temporal Classification of Urban Crime Types in the United Kingdom Using Supervised Machine Learning

Masters in Data Science & AI • Metropolitan Police Service Open Data • COMP7090 Research Methods

Problem & Aim

Crime in the UK remains a critical concern — with a **4.1% increase** in Q4 2025 and a reported rate of **74 per 1,000 people**. Manually identifying patterns in large volumes of street-level crime data is time-intensive and error-prone.

Aim: To develop, evaluate, and interpret supervised machine learning models for **multi-class classification of urban crime types** using open Metropolitan Police data — and to assess validity, limitations, and practical implications for proactive policing.

Task: Given *when* (temporal features) and *where* (spatial features) a crime occurred, predict *what type* of crime it was across 14 categories.

Scope & Dataset

- Source:** UK Police Open Data Portal (data.police.uk)
- Force:** Metropolitan Police Service (London)
- Period:** Configurable — default 12 months (Jan–Dec 2025)
- Records:** ~955,000 street-level crime records after cleaning
- Target:** 14 crime categories (e.g. Burglary, Shoplifting, Violence and sexual offences, Robbery)
- Features:** Temporal (cyclical month, quarter, season) + Spatial (LSOA aggregation, raw lat/lon)
- Licence:** Open Government Licence — publicly accessible for research

Hypotheses

H1 Supervised ML models can classify crime type significantly better than random and majority-class baselines.

PARTIALLY SUPPORTED — tree-based models only

H2 Spatial aggregation (LSOA/grid) improves classification vs raw latitude/longitude.

FALSIFIED — raw lat/lon outperformed for RF (+0.05 F1)

H3 Class imbalance mitigation improves recall for minority crime classes without large precision loss.

FALSIFIED — built-in class weighting outperformed SMOTE/ROS

Data Science & AI Workflow

- Data Acquisition**
Load monthly CSVs from UK Police zip archives via configurable multi-month loader. 12 months of MPS street-level crime records.
- Cleaning & Preprocessing**
Drop rows with missing essentials, parse dates, filter invalid coordinates. Standardise column names via auto-detection.
- Feature Engineering**
Derive cyclical month encoding (sin/cos), squared month, quarter, season. Spatial: frequency-based LSOA collapsing with rare-bin bucketing.
- Train-Test Split**
Temporal 80/20 split (months 1–10 train, 11–12 test). Sentinel remapping for unseen spatial codes in test set.
- Baseline Computation**
Random baseline and majority-class baseline (H1 floor).
- Model Training**
Four model families across two paradigms:
Classical ML: Logistic Regression (scaled), Random Forest (100 trees, balanced)
Boosted Trees: XGBoost (sample-weighted for class balance)
Deep Learning: MLP — 3-layer feedforward neural network (128→64→32)
- H2 Experiment**
Compare raw lat/lon vs aggregated LSOA features for LR and RF.
- H3 Experiment**
Compare SMOTE, RandomOverSampler, and built-in class weighting.
- Evaluation**
Macro F1, per-class precision/recall/F1, confusion matrices, feature importance, SHAP.

Tools & Technologies

- Environment:** Google Colab (T4 GPU)
- Language:** Python 3.11
- Data handling:** pandas, NumPy
- Classical ML:** scikit-learn (LR, RF, StandardScaler, metrics)
- Boosted trees:** XGBoost
- Deep learning:** PyTorch (MLP, CUDA acceleration)
- Imbalance handling:** imbalanced-learn (SMOTE, RandomOverSampler)
- Interpretability:** SHAP, RF feature importance
- Visualisation:** matplotlib, seaborn

Key Results — Macro F1 Scores

0.060 Random Baseline	0.028 Majority Baseline
0.029 Below Baseline	0.137 Best Model
0.094 Better Baselines	0.062 Near Random
0.187 Raw Lat/Lon	0.137 Aggregated LSOA
0.137 Balanced Subsample	0.100 SMOTE
	0.117 Random OverSampler

H2 — Spatial Representation Comparison (RF)

H3 — Class Imbalance Strategies (RF)

Evaluation & Impact

Key finding: Random Forest is the strongest model (macro F1 = 0.137), confirming that temporal and spatial features carry **non-linear signal** that tree-based models can exploit but linear models and neural networks cannot — given integer-encoded categorical inputs.

Critical insight: LabelEncoder on LSOA codes destroys spatial proximity — raw lat/lon outperformed aggregated features by +0.05 F1. The encoding method matters more than the aggregation strategy itself.

Practical implication: With only temporal and spatial features, crime type prediction remains inherently limited. The features that distinguish crime types (victim, premises, offender behaviour) are absent from the open dataset. This ceiling, not model choice, is the primary constraint.

Future work: Integrate ONS socio-economic covariates (Index of Multiple Deprivation) by LSOA, which may substantially improve predictions by providing area-level context that correlates with crime type distribution.

Ethical Considerations

- Reporting bias:** Police data reflects reported/recorded crime, not actual crime — over-policed areas appear to have higher crime rates
- Fairness risk:** Spatial features may encode socio-economic or demographic proxies, risking discriminatory resource allocation
- Data licence:** Open Government Licence — no personal identifiers in dataset